



Enhanced electrical properties in A-site K/Ce and B-site W/Cr co-substituted $\text{CaBi}_2\text{Nb}_2\text{O}_9$ high temperature piezoelectric ceramic

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Abstract

A-site K/Ce and B-site W/Cr co-substituted $\text{CaBi}_2\text{Nb}_2\text{O}_9$ ceramics were synthesized by a conventional solid-state reaction process. The structures, piezoelectricity and electrical conduction behaviors were investigated in detail. The $\text{Ca}_{0.95}(\text{K}_{1/2}\text{Ce}_{1/2})_{0.05}\text{Bi}_2\text{Nb}_{1.99}(\text{W}_{2/3}\text{Cr}_{1/3})_{0.01}\text{O}_9$ ceramic exhibited best properties with a high d_{33} (piezoelectric coefficient) of ~ 18.4 pC/N and a high T_C (Curie temperature) of ~ 917 °C. In addition, the ceramic displayed an excellent thermal stability performance such that d_{33} remained 94% of its initial value even after annealing at 900 °C for 2 h. The planar electro-mechanical coupling factor was observed to increase from 9.00% at room temperature to 15.24% at 600 °C. Furthermore, a high electrical resistivity of 1.16×10^5 Ω cm at 600 °C and a good fatigue property were also achieved with this composition. The $\text{Ca}_{0.95}(\text{K}_{1/2}\text{Ce}_{1/2})_{0.05}\text{Bi}_2\text{Nb}_{1.99}(\text{W}_{2/3}\text{Cr}_{1/3})_{0.01}\text{O}_9$ ceramic was found to be a promising candidate for high-temperature piezoelectric applications.

1 Introduction

Due to environmental concerns arising from the toxicity of lead based compounds, there is a need for exploring lead-free ceramics for piezoelectric applications. Among the existing lead-free piezoelectric ceramics, Aurivillius phase ferroelectrics with bismuth-layered perovskite structure (BLSFs) have received significant attention due to their high Curie temperatures and excellent fatigue properties [1–3]. This has enabled their applicability in a wide range of industrial and scientific fields such as high temperature piezoelectric transducers and nuclear power industries [4]. The structure of BLSFs consists of pseudo-perovskite $(\text{A}_{n-1}\text{B}_n\text{O}_{3n+1})^{2-}$ and $(\text{Bi}_2\text{O}_2)^{2+}$ layers, where A is a series of cations, such as Ca^{2+} , Sr^{2+} , Ba^{2+} , K^+ , rare earth ions and so on, B includes some small highly charged cations like Nb^{5+} ,

Ta^{5+} , W^{6+} , and n is the number of octahedral layers in the perovskite layers which can range from 1–6 layers [5, 6].

As an important member of the BLSFs, $\text{CaBi}_2\text{Nb}_2\text{O}_9$ (CBN) ceramics possesses an ultra-high Curie temperature (T_C) of about 940 °C [2], making it a potential candidate for high temperature applications above 600 °C. However, the piezoelectric activity of CBN ceramic is very poor (~ 5 pC/N) owing to the higher conductivity in the *a-b* plane as compared to the *c* axis direction, which makes the poling difficult [7]. Several approaches have been proposed in order to improve the piezoelectric activity of CBN ceramics [2, 8, 9]. Among them, A-site co-substitution is regarded as a practical approach to enhance the piezoelectric properties of CBN. With diatomic doping (X, Ce) at A-site, CBN-based ceramics exhibit an enhanced d_{33} (piezoelectric coefficient) from 5 pC/N of pure CBN to 13.3, 16, and 16 pC/N for X = Li, K, Na, respectively [10–12]. Moreover, on the basis of (Li, Ce) co-doping, triatomic Li, Ce, and A co-doped CBN ceramics with A = Y, Nd, and Pr show d_{33} values of 16.1, 13.1, and 17.3 pC/N, respectively [13–15]. Similarly, enhancement of the piezoelectric properties have also been achieved in B-site substituted CBN ceramics. For instance, Xing et al. reported Zr-doped CBN ceramic with a high d_{33} of ~ 11.6 pC/N [16]. In our previous work, W/Cr co-doped CBN ceramic with high d_{33} (15 pC/N) and excellent resistivity (4.9×10^5 Ω cm at 600 °C) was obtained [17]. Although

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several improvements have been demonstrated, the piezoelectric properties of CBN-based ceramics is still lower than that of other BLSFs [18–20], thereby warranting further studies.

In recent years, a complex doping strategy that simultaneously substituted the A and B-site atoms with different elements has been proposed to improve the piezoelectric activity of BLSFs ceramics. Large enhancement of d_{33} was achieved in co-substituted $\text{Na}_{0.5}\text{Bi}_{2.5}\text{Nb}_2\text{O}_9$ ceramics with Li/Ce at A-site and W at B-site [21]. The d_{33} value of CBT-based ceramics also increased from 7 to 19.6 pC/N with co-doping of Li/Ce at A-site and Nb/Mn at B-site [22]. Long et al. reported a co-substituted CBN-based ceramic with Li/Nd on the A-site and Sc/W on the B-site with a remarkable d_{33} value of 20.6 pC/N, which is among the highest values reported for CBN-based ceramics [23]. These studies demonstrated the efficacy of the complex doping strategy in improving the piezoelectric properties of BLSFs ceramics. In spite of the potential of simultaneous A and B-site substituted CBN-based ceramics, only a few studies have been reported. In this study, we selected A-site K/Ce and B-site W/Cr co-modification to improve the piezoelectric properties of CBN ceramic, in which the K/Ce or W/Cr has been independently proven to be effective. The structures, piezoelectric properties, ferroelectric properties and electrical properties were studied in detail.

2 Experimental methods

The $\text{Ca}_{0.95}(\text{K}_{1/2}\text{Ce}_{1/2})_{0.05}\text{Bi}_2\text{Nb}_{1.99}(\text{W}_{2/3}\text{Cr}_{1/3})_{0.01}\text{O}_9$ (CBN-KCe-WCr) ceramic was prepared using a solid-state reaction technique. CaCO_3 (99.99%), Bi_2O_3 (99.99%), Nb_2O_5 (99.5%), K_2CO_3 (99.5%), CeO_2 (99.99%), WO_3 (99.8%), Cr_2O_3 (99.95%) were used as raw materials and weighed according to the stoichiometric compositions. All powders

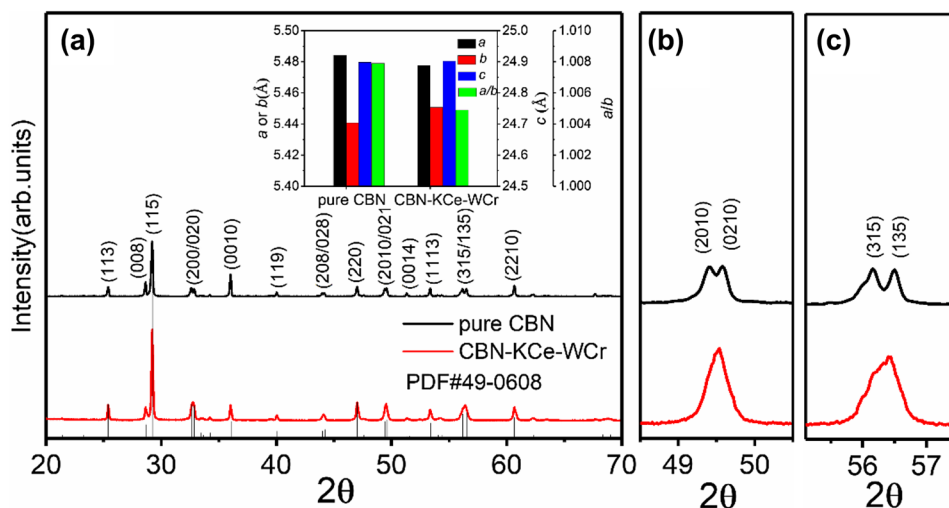
were mixed in ethanol in a ball mill for 12 h using a zirconia grinding media and then the dried mixtures were calcined at 750 °C for 3 h. The calcined powder was milled again under the same conditions and pressed into pellets (12 mm in diameter, 1 mm in thickness) under a pressure of around 200 Mpa using polyvinyl alcohol (PVA) as a binder. After burning out PVA at 550 °C, the pellets were sintered at 1175 °C for 3 h. For the electrical measurements, silver paste was painted on both surfaces of the sintered ceramics and fired at 825 °C for 20 min in the air atmosphere. The poled disks were annealed for 2 h from room temperature to 1000 °C and the d_{33} values were measured at room temperature.

The crystalline phases of the samples were determined by X-ray diffraction (XRD, Ultima IV). The natural surface of the CBNWC-x samples was examined by scanning electron microscopy (SEM, JEOL JSM6460-LV). The dielectric properties of the samples were obtained using an impedance analyzer (Keysight E4990A). The resistivity was measured using a high-resistance meter (Keithley 2410). Ferroelectric hysteresis loops, the strain curves and fatigue tests of the samples were performed using a ferroelectric analyzer (TF200, AixACCT, Aachen). The samples were poled under a DC (direct current) electric field of 13–15 kV/mm in silicone oil at 160 °C for 30 min. d_{33} of the samples were measured at room temperature using a piezo- d_{33} meter (ZJ-3AN).

3 Results and discussion

Figure 1a shows the XRD patterns of pure CBN and CBN-KCe-WCr ceramics in the 2θ range of 20–70°. All the diffraction peaks can be indexed by the JCPDS 49-0608 database. No secondary phase was detected, suggesting that K/Ce and W/Cr ions have diffused into the CBN lattice. The strongest diffraction peak was (115), in accordance with the fact that the most intense reflection of BLSFs is the type of

Fig. 1 **a** XRD patterns of pure CBN and CBN-KCe-WCr ceramics, with an inset of Lattice parameters a , b , c and a/b of all ceramics. **b** Enlarged view of (2010/0210) diffraction peaks. **c** Enlarged view of (315/135) diffraction peaks



(112 $m + 1$) [24, 25]. Enlarged views of (2010/0210) and (315/135) diffraction peaks are shown in Fig. 1b, c, respectively. Notably, these two peaks had a tendency to merge into one peak with the K/Ce and W/Cr doping, indicating the occurrence of a structural distortion [26].

To further identify the phase structure of this material system, the lattice parameters a , b and c of pure CBN and CBN-KCe-WCr ceramics were calculated, as shown in the inset of Fig. 1a.

The lattice parameters b and c were found to increase while a decreased as K/Ce and W/Cr ions were doped into CBN. The a/b ratio was usually defined as tetragonality [27]. Since the a/b ratio decreased with the introduction of K/Ce and W/Cr dopants, it led to a transition from an orthorhombic symmetry to tetragonal symmetry. Strengthening of the tetragonal distortion is favorable to the polarization switching, and maintains a high theoretical P_s of ceramics, thus realizing an enhancement of d_{33} [27].

The SEM images of natural surface of the CBN ceramics were characterized, as shown in Fig. 2a, b. Both pure CBN and CBN-KCe-WCr ceramics exhibited typical plate-like grains, which is a characteristic of Aurivillius ceramics. The CBN-KCe-WCr ceramic showed relatively uniform morphology compared with the pure CBN ceramic. It might be ascribed to the W/Cr co-doping at B-sites, which suppressed the anisotropic growth of grain and leads to fine isotropic grains distributions [17, 28]. Elemental mapping of the natural surface of CBN-KCe-WCr ceramic is shown in Fig. 2c–k. The black regions in the mapping patterns may be caused by the uneven surface, as the black regions always appear in the junction of grains where is out of flatness. A

homogenous distribution of elements can be still observed inside the grain, which confirmed the diffusion of K/Ce and W/Cr ions into the CBN lattice.

Figure 3 shows the temperature dependence of dielectric constant and loss tangent of pure CBN and CBN-KCe-WCr ceramics at 1 MHz. From the inset of Fig. 3a, it is evident that T_C decreased from 943 °C to 917 °C after the introduction of K/Ce and W/Cr. Generally, the transition temperature (T_C) from ferroelectric to paraelectric phase is closely related the orthorhombic distortion of BLSFs [23]. The tolerance factor (t) of the ceramics was given by the following formula:

$$t = \frac{(r_A + r_O)}{\sqrt{2}(r_B + r_O)} \quad (1)$$

where, r_A , r_B , and r_O are the ionic radii of the A-site cations, B-site cations, and oxygen ions, respectively [29, 30]. Lower t value always corresponds to higher T_C , which has been examined in several earlier studies [13, 31]. Considering that the ionic radii of K^+/Ce^{3+} (0.138 nm/0.102 nm) were larger than that of the Ca^{2+} (0.100 nm) while the ionic radii of W^{6+} (~0.060 nm) and Cr^{3+} (~0.052 nm) were smaller than that of Nb^{5+} (~0.064 nm) [20, 32–34], a higher t can be expected, which may be responsible for the decrease of T_C . Moreover, it was found from Fig. 3b that though the dielectric loss of CBN-KCe-WCr ceramic was higher than that of pure CBN ceramic, it was still very low (< 5%) when the measurement temperature was below 500 °C.

Figure 4a shows the d_{33} values of the pure CBN and CBN-KCe-WCr ceramics, where the d_{33} value of pure CBN

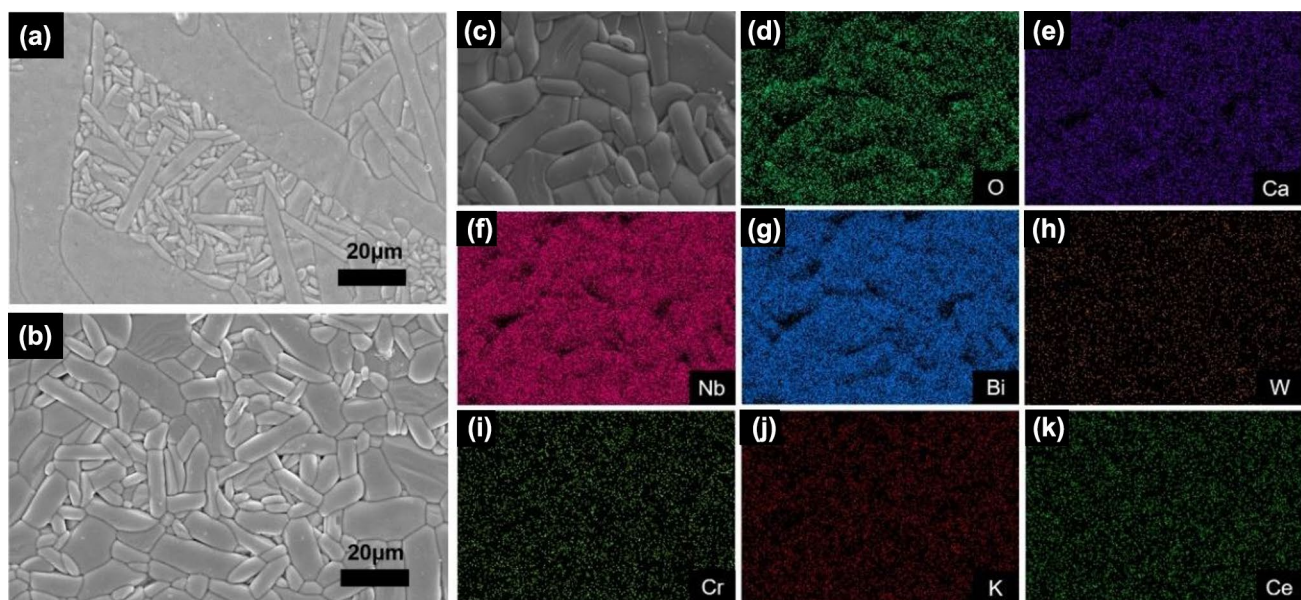


Fig. 2 SEM images of natural surface of the ceramics. **a** Pure CBN **b** CBN-KCe-WCr **c–k** Elemental mapping of the CBN-KCe-WCr ceramic

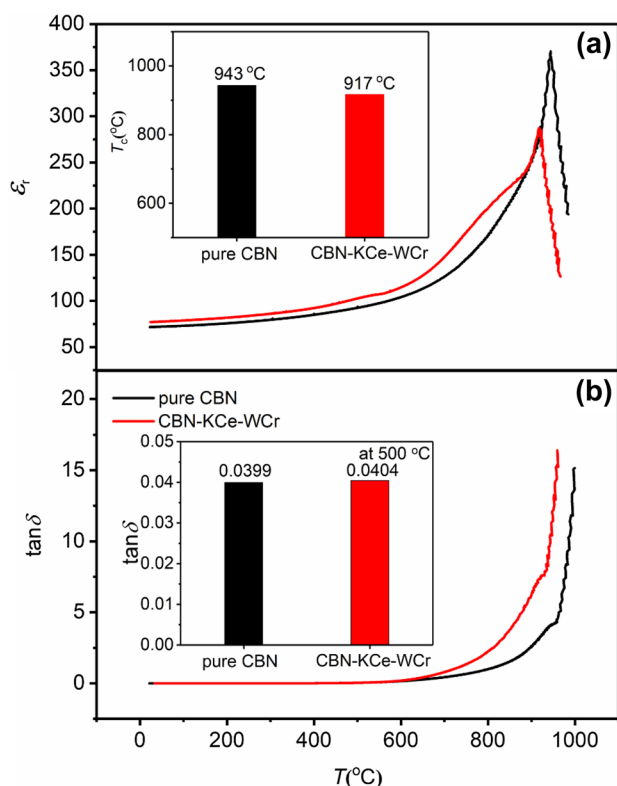


Fig. 3 Temperature dependence of dielectric permittivity ϵ_r and dielectric loss $\tan\delta$ of pure CBN and CBN-KCe-WCr ceramics at 1 MHz. **a** Dielectric permittivity versus Temperature, with an inset showing the change trends in T_c . **b** Dielectric loss versus Temperature, with an inset showing the $\tan\delta$ at 500 $^{\circ}\text{C}$

ceramic was found to be 5 pC/N, which is consistent with the previous work reported by Yan et al. [2]. With K/Ce and W/Cr co-substitutions, the d_{33} value was significantly enhanced to 18.4 pC/N. Since this d_{33} value was higher than the individual values of 16 pC/N for A-site K/Ce or 15 pC/N for B-site W/Cr doped CBN ceramics, it suggested that the combined substitution of K/Ce and W/Cr was more advantageous than individual substitutions. A summary of literature data relating to the correlation of T_c and d_{33} of CBN ceramics is shown in Fig. 4b [10, 12–15, 17, 23, 27, 35–39]. Generally, the enhancement of d_{33} produced by doping was usually accompanied by the decrease of T_c . For the CBN-based ceramics, most of the study results located in the green strip region. Obviously, the result of this work (red star) lie outside of the green strip region, demonstrating that it has transcended most of the reported CBN-based ceramics.

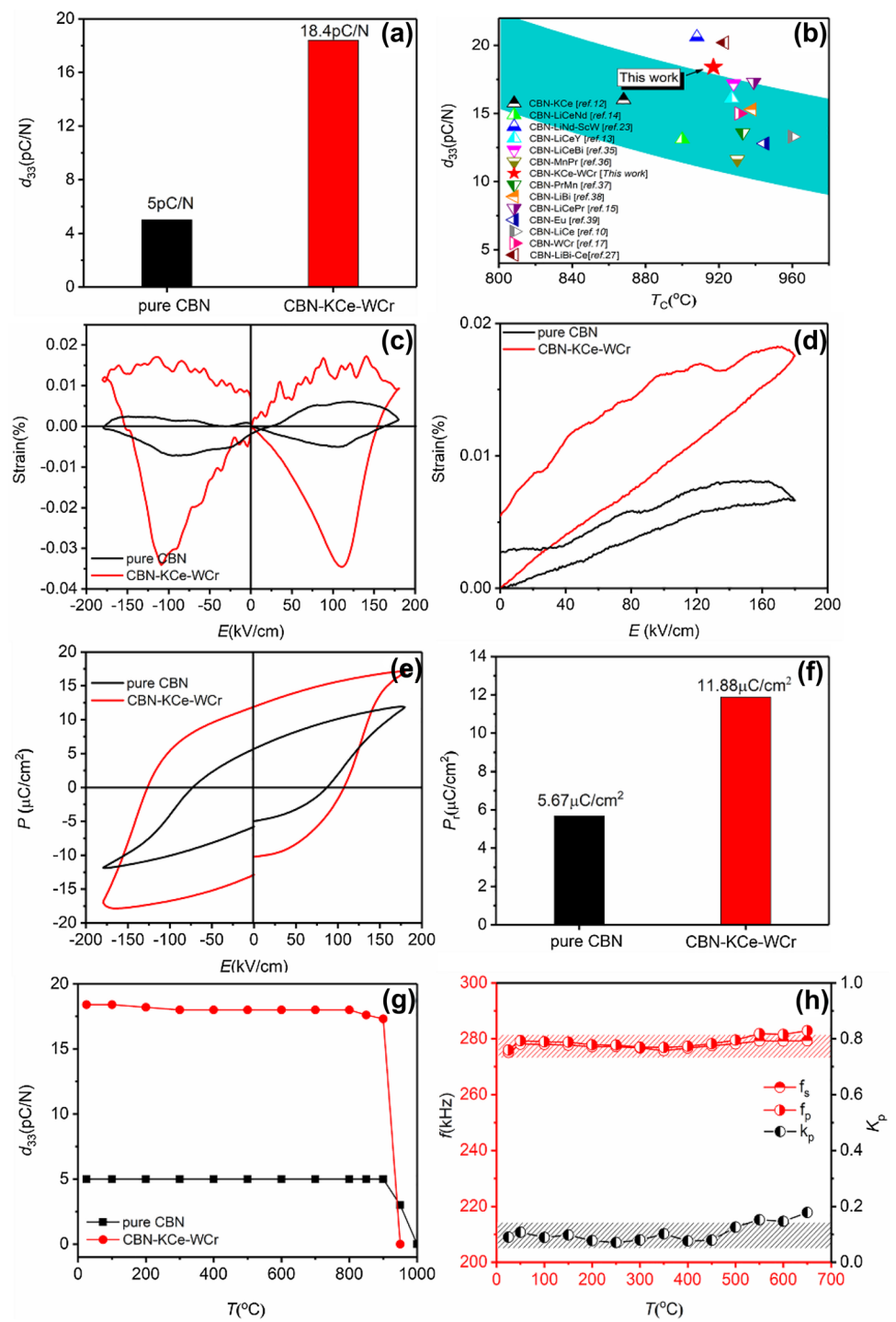
Figure 4c, d shows the bipolar and unipolar field-induced strains of pure CBN and CBN-KCe-WCr ceramics and the unipolar strains, respectively. The maximum unipolar strain of CBN-KCe-WCr ceramic was 0.018%, which was thrice that of the pure CBN ceramic (0.006%). Similar enhancement can be observed in the bipolar strain. Figure 4e shows

the polarization hysteresis loops of the ceramics with a fixed field of 180 kV/cm. Both ceramics exhibited saturated hysteresis loops. The obtained remanent polarization of the samples is shown in Fig. 4f, where the CBN-KCe-WCr ceramic had a P_r of 11.88 $\mu\text{C}/\text{cm}^2$, about twice that of pure CBN (5.67 $\mu\text{C}/\text{cm}^2$), which could be partly responsible for the enhancement of d_{33} . The remarkable enhancement of P_r may be ascribed to the decrease of a/b ratio as evidenced by the XRD results, since the spontaneous polarization switching of CBN is limited in the a – b plane. Additionally, the optimization of microstructure could be another reason for the enhancement of d_{33} and P_r .

In order to investigate the piezoelectric properties of the co-substituted ceramics, their thermal stability was evaluated. The effect of thermal depoling on the variation of d_{33} for the pure and co-substituted ceramics is shown in Fig. 4g. Both pure CBN and CBN-KCe-WCr ceramics showed nearly temperature independent d_{33} when the annealing temperature was below 900 $^{\circ}\text{C}$. While the d_{33} values dropped rapidly and finally became almost zero when the temperature was raised beyond the T_c . It should be noted that d_{33} value of pure CBN did not become zero after annealing at 950 $^{\circ}\text{C}$, which possessed a T_c value of 943 $^{\circ}\text{C}$. Similar phenomenon has also been observed in previous reports [40–42]. It might be ascribed to the pinning effect induced by the higher oxygen vacancies concentration, which restricted the reverse switch of the domain during the thermal depoling [13]. Notably, the CBN-KCe-WCr ceramic still had a high d_{33} value of 17.3 pC/N after thermal depoling at 900 $^{\circ}\text{C}$, indicating that a stable piezoelectric property as a function of temperature. In addition, the resonance frequencies f_s , f_p and planar electromechanical coupling factor k_p versus temperature of the CBN-KCe-WCr ceramic are shown in Fig. 4h. f_s and f_p were found to be close to 280 kHz over the entire temperature range. k_p increased from 9.00% at room temperature to 15.24% at 600 $^{\circ}\text{C}$, again indicating stable piezoelectric properties.

Figure 5a shows DC resistivity as a function of temperature for pure CBN and CBN-KCe-WCr ceramics in the temperature range of 200–800 $^{\circ}\text{C}$. A three-fold improvement was observed in the resistivity of CBN-KCe-WCr ceramic ($1.16 \times 10^5 \Omega \text{ cm}$) before 600 $^{\circ}\text{C}$ as compared to undoped CBN ($4.49 \times 10^4 \Omega \text{ cm}$), demonstrating that the K/Ce and W/Cr co-doping was beneficial towards the enhancement of resistivity. For temperatures higher than 600 $^{\circ}\text{C}$, the resistivity of the two samples was similar, suggesting that the intrinsic contribution was predominant at high temperatures. A similar phenomenon has also been reported in W donor-doped $\text{Bi}_3\text{TiNbO}_9$ ceramics [43]. A summary of d_{33} and ρ at 600 $^{\circ}\text{C}$ of CBN-based and BTNO-based ceramics with T_c around 900 $^{\circ}\text{C}$ derived from previous reports along with those obtained in this work is shown in Fig. 5b [13, 14, 17, 27, 37, 39, 44]. Compared with those CBN-based and

Fig. 4 **a** d_{33} values of the pure CBN and CBN-KCe-WCr ceramics. **b** A summary of literature data about the correlation of T_C and d_{33} of CBN ceramics. **c** Bipolar strain curves. **d** Unipolar strain curves. **e** Polarization hysteresis loops. **f** P_r values of the pure CBN and CBN-KCe-WCr ceramics. **g** The effect of thermal depoling on the change of d_{33} after annealing for 2 h. **h** Resonance frequency f_s , f_p and planar electromechanical coupling factor k_p versus Temperature of CBN-KCe-WCr ceramic



BTNO-based ceramics, the CBN-KCe-WCr ceramic exhibited a high d_{33} along with a relative high resistivity, thereby establishing that CBN-KCe-WCr ceramic is suitable for high-temperature piezoelectric applications.

To elucidate the mechanism for the enhancement of DC resistivity, (AC) alternating current conduction behaviors of pure CBN and CBN-KCe-WCr ceramics were measured from 100 Hz to 1 MHz at various temperatures, as shown in

Fig. 5c, d. Both the pure CBN and CBN-KCe-WCr ceramics exhibited a single arc, indicating that only the grain contributed to the conductivities of the ceramics [45]. In addition, the decreased diameter of arc upon the increase of temperature could also be observed, which can be attributed to the lower resistance. A model of a parallel equivalent circuit (R-CPE-C) was established to fit the resistance using Z-view software, as shown in the inset of Fig. 5. The impedance

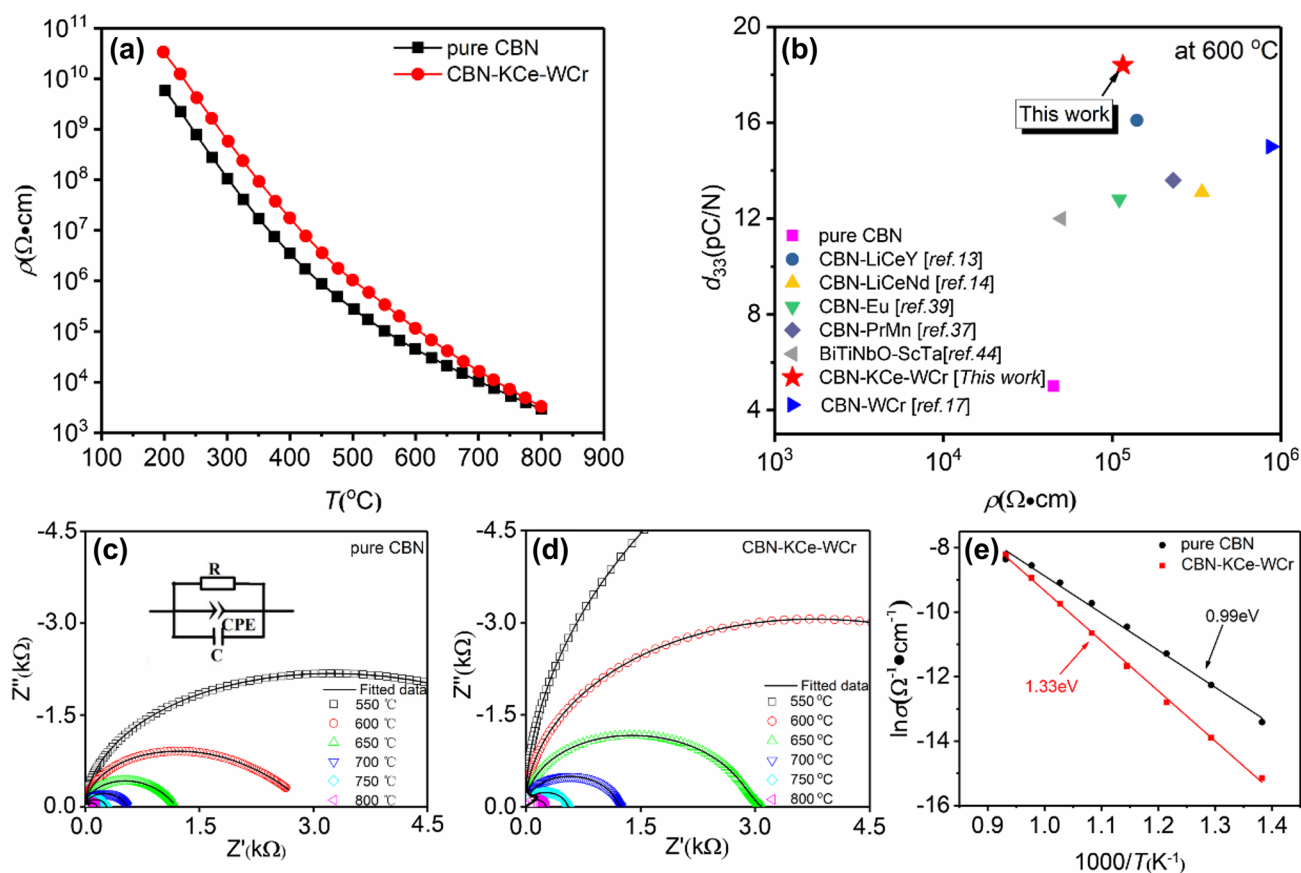


Fig. 5 **a** Variation of the resistivity of CBNWC-x ceramics from 200 °C to 800 °C. **b** Comparison of d_{33} and ρ at 600 °C of CBN-based and BTNO-based ceramics derived from the previously published work and those in this work. **c, d** Cole–Cole plots of imped-

ance for pure CBN and CBN-KCe-WCr ceramics measured at various temperatures, respectively. **e** Arrhenius relationship between conductivity and temperature

plots of both the samples fitted well with the experimental data. The activation energies E_a can be calculated by the Arrhenius law:

$$\sigma = A \exp(-E_a/kT) \quad (2)$$

where, A is the pre-exponential factor, E_a is the activation energy of carrier migration, and k is the Boltzmann constant. Figure 5e plotted the $\ln \sigma$ as a function of $1000/T$. The corresponding E_a of the pure CBN was determined to be 0.99 eV, indicating that the electrical conduction process of the undoped CBN ceramic was dominated by oxygen vacancies [46, 47]. The doped CBN ceramic showed a higher E_a value of ~1.33 eV, which suggests that K/Ce and W/Cr doping can effectively decrease the concentration of oxygen vacancies and lead to the enhanced DC resistivity shown in Fig. 5a. The mechanism for the decrease of oxygen vacancies in the co-doped ceramic is still unclear and two reasons are speculated to account for that. On the one hand, the W^{6+} substituted for Nb^{5+} could accompany by eliminating of oxygen vacancies in order to achieve charge neutrality [48]. On the other hand, the doped Cr^{3+} for Nb^{5+}

might form defect dipoles ($Cr_{Nb}''-V_O^{\bullet\bullet}$), which could reduce the concentration of oxygen vacancies neighbouring Nb^{5+} ions [49]. Further studies are needed to ascertain the exact cause of the lowering of oxygen vacancies.

The decrease of oxygen vacancies also has an effect on fatigue properties of ceramics, since the oxygen vacancies tend to agglomerate on the domain walls and finally pin them. Figure 6 shows the variation of ferroelectricity as a function of the number of switched cycles. It was observed that beyond 10^3 cycles of bipolar fatigue, the P_r of pure CBN ceramic had an obvious enhancement, which could be ascribed to the pinning effect relaxation in the domain walls owing to the repeated polarization reversal [23]. The P_r value of 10^5 cycles was two and a half times larger than that of the initial value. This large increase in P_r value also demonstrates that large pinning effect exists in the undoped CBN ceramic, which is attributed to the high oxygen vacancy concentration. On the contrary, the CBN-KCe-WCr ceramic exhibited a good fatigue stability with no apparent change below 10^4 cycles, which may be due to fewer oxygen vacancies in CBN-KCe-WCr ceramic,

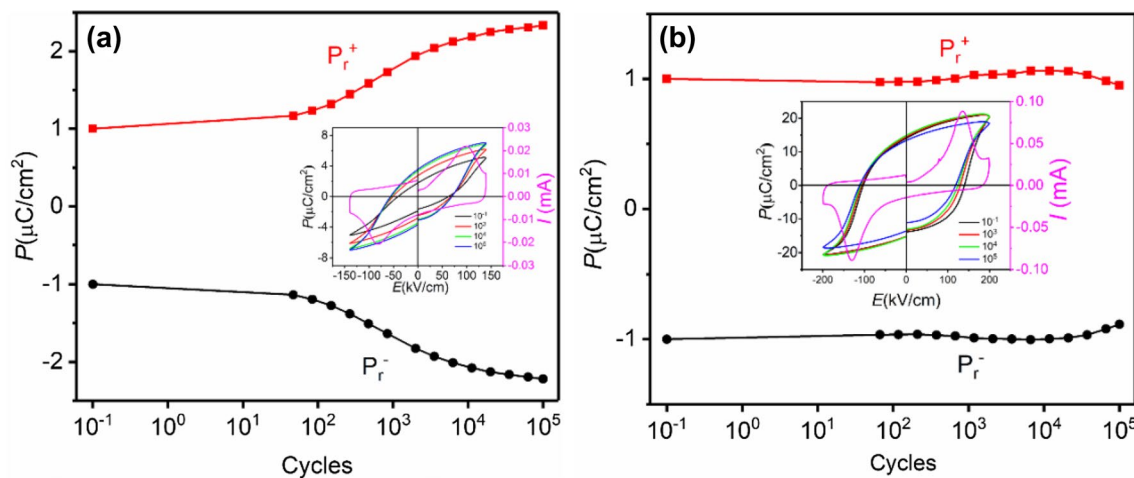


Fig. 6 **a, b** Variation of ferroelectricity versus number of switched cycles of pure CBN and CBN-KCe-WCr ceramics, respectively, with an inset showing compared P - E loops after different switched cycles

where the pinning effect is much weaker as compared to pure CBN. However, when the switching cycles increased to 10^5 times, an obvious degradation of P_r was observed in CBN-KCe-WCr ceramic, suggesting the agglomeration of small amount of oxygen vacancies on the domain walls [23, 50].

4 Conclusions

A modified ultrahigh-temperature piezoelectric ceramic $\text{Ca}_{0.95}(\text{K}_{1/2}\text{Ce}_{1/2})_{0.05}\text{Bi}_2\text{Nb}_{1.99}(\text{W}_{2/3}\text{Cr}_{1/3})_{0.01}\text{O}_9$ was prepared by a conventional solid-state reaction process, where K/Ce co-substituted the A-site Ca and W/Cr co-substituted B-site Nb, respectively. The modification by K/Ce and W/Cr altered the piezoelectric and ferroelectric properties of CBN ceramic, due to the structural distortion, microstructure optimization and reduction of oxygen vacancy concentration. A high d_{33} value of 18.4 pC/N, high T_C of 917 °C, high remanent polarization of 11.88 $\mu\text{C}/\text{cm}^2$ and a large field-induced strain of 0.018% were observed in CBN-KCe-WCr ceramic. Additionally, an excellent thermal stability of 17.3 pC/N was also obtained in the ceramic, which persisted even after annealing at 900 °C. This work demonstrates that A-site and B-site co-modification can lead to better piezoelectric properties as compared to individual A-site or B-site modifications and the CBN-KCe-WCr ceramic can be suitable for high-temperature piezoelectric applications.

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